

Telco Customer Churn - Complete Jupyter Notebook

1. IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use('default')
```

2. LOAD DATASET

```
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn_Sumasri.csv")
df.head()
```

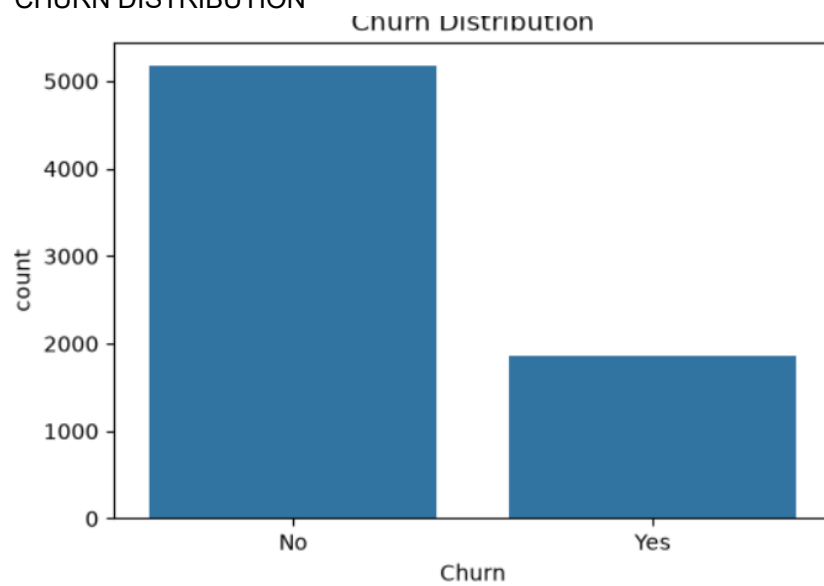
3. BASIC INFO

```
df.shape
df.info()
df.describe()
```

4. DATA CLEANING

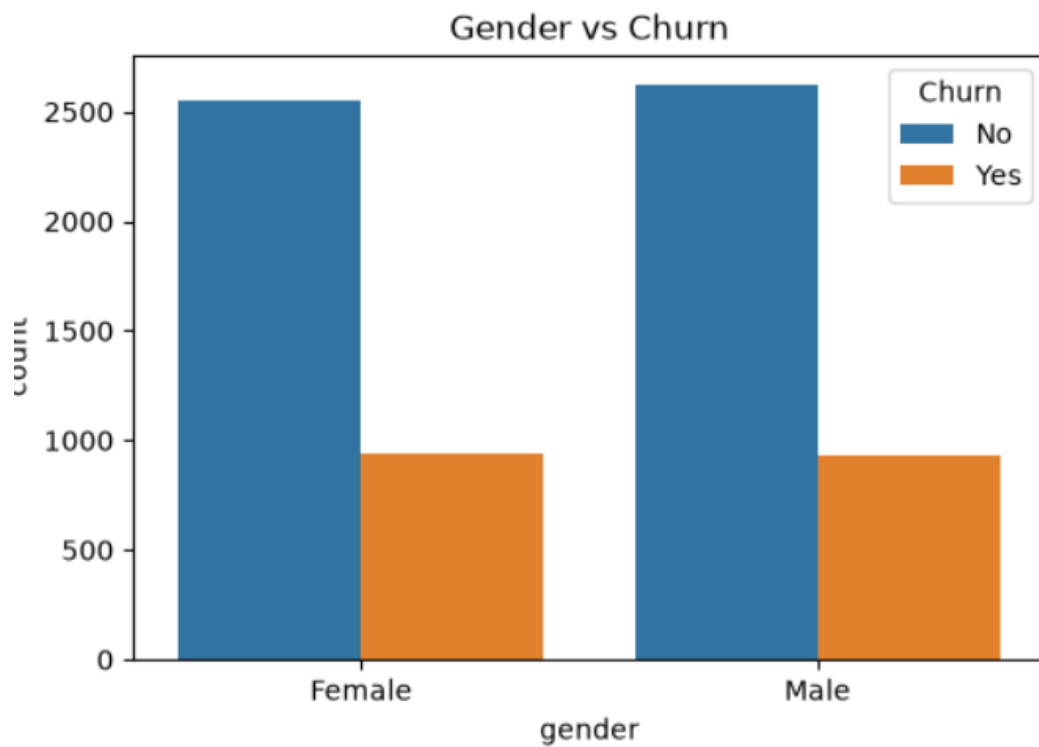
```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors='coerce')
df["TotalCharges"].fillna(df["TotalCharges"].median(), inplace=True)
```

5. CHURN DISTRIBUTION



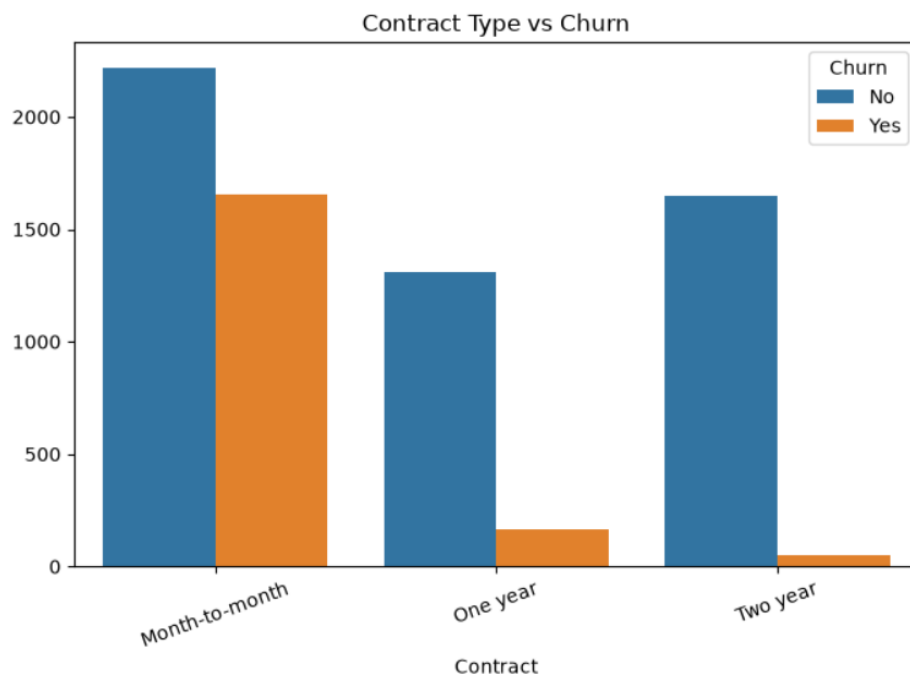
```
sns.countplot(x='Churn', data=df)
```

6. GENDER VS CHURN



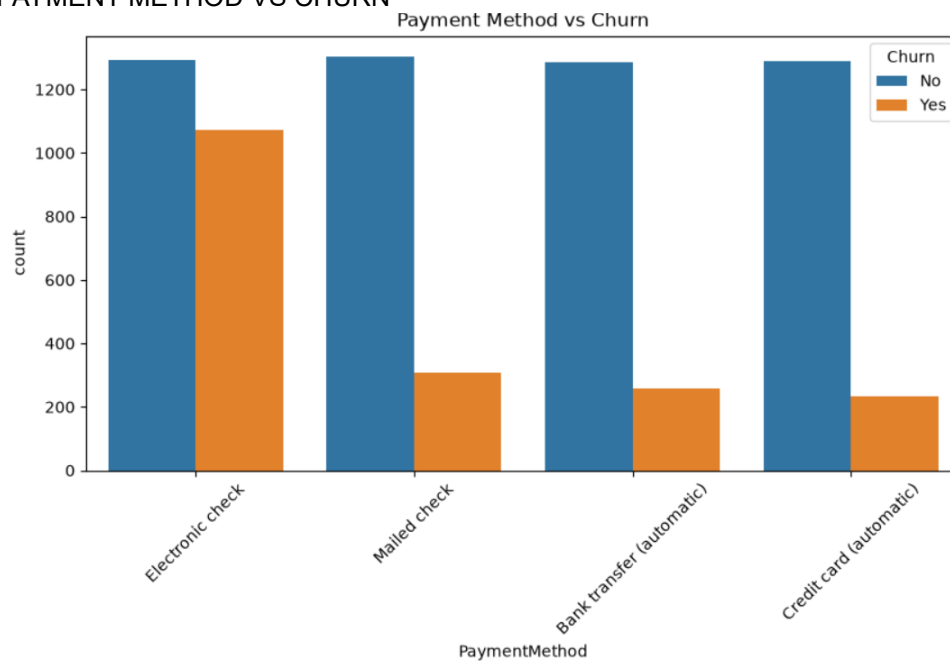
```
sns.countplot(x='gender', hue='Churn', data=df)
```

7. CONTRACT VS CHURN



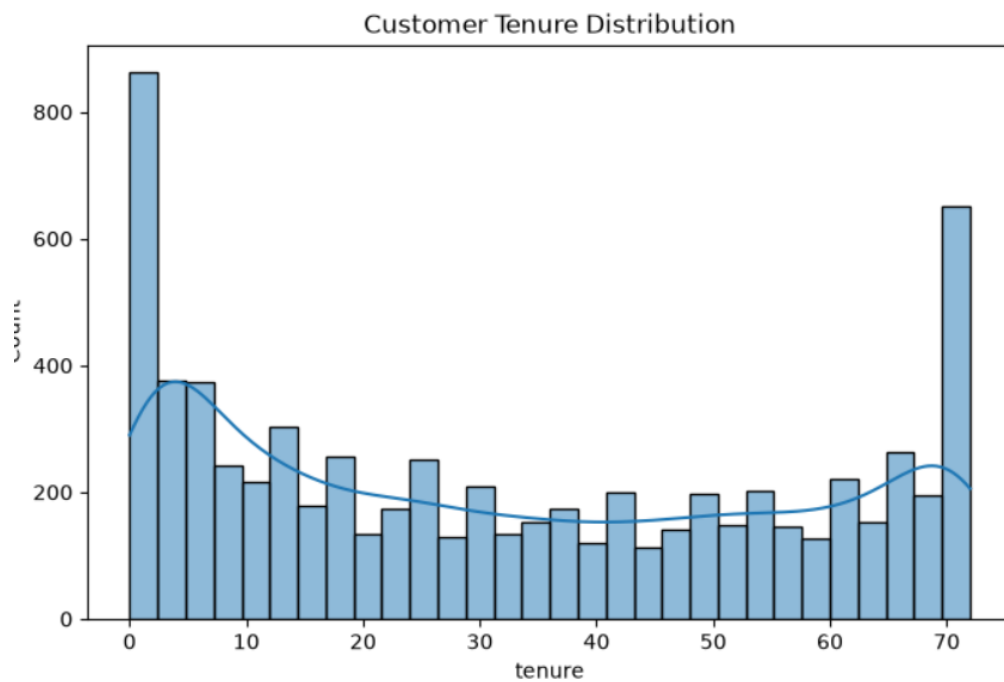
```
sns.countplot(x='Contract', hue='Churn', data=df)
```

8. PAYMENT METHOD VS CHURN



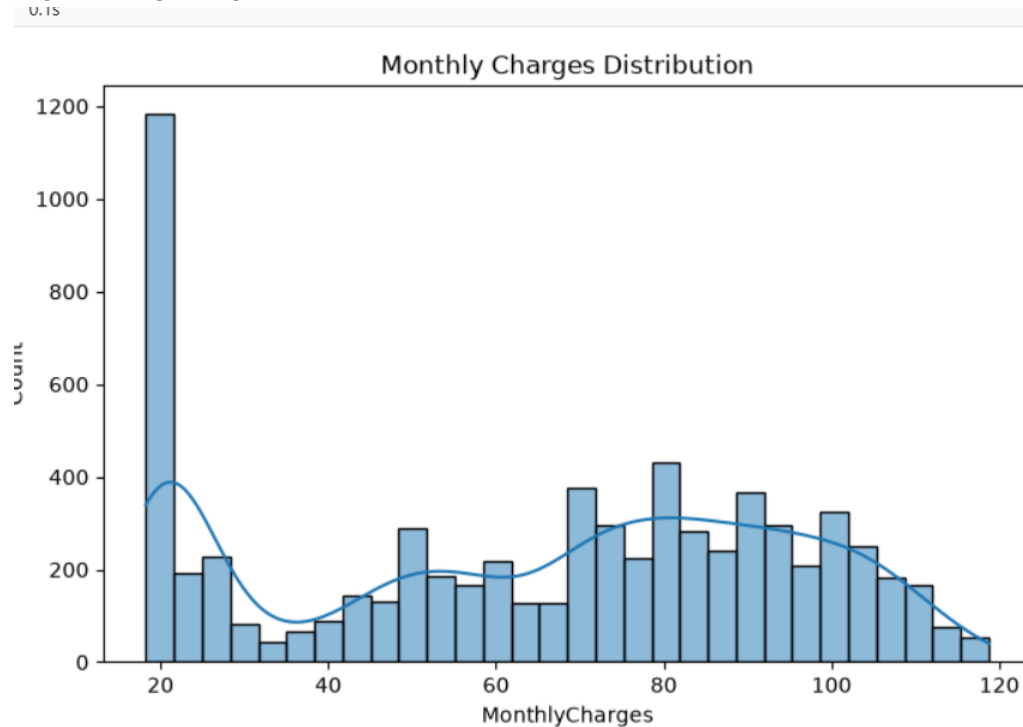
```
sns.countplot(x='PaymentMethod', hue='Churn', data=df)
```

9. TENURE DISTRIBUTION



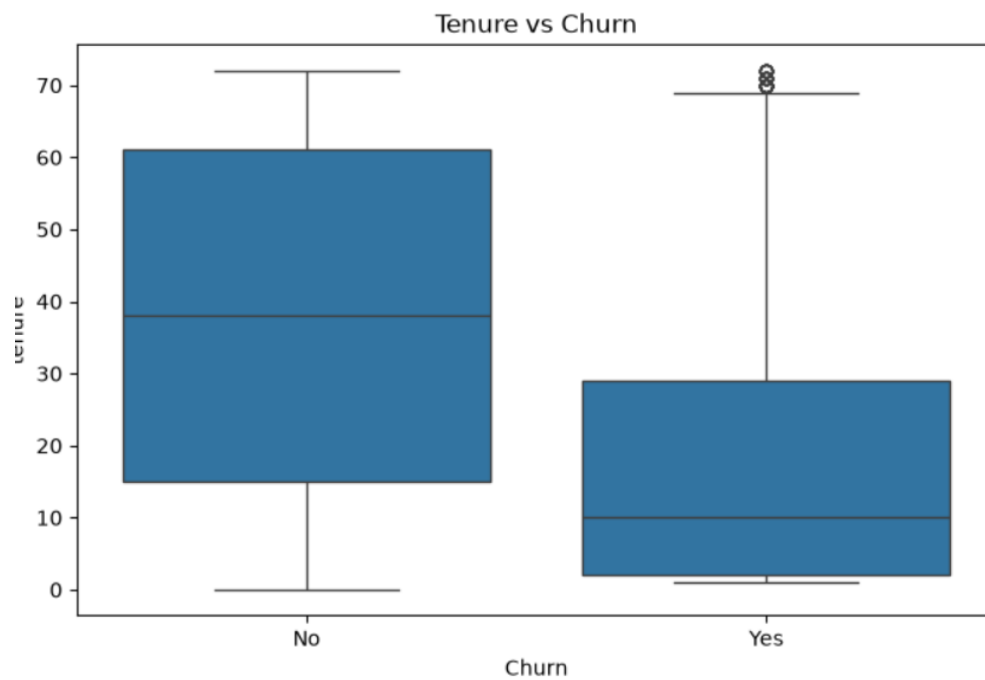
```
sns.histplot(df['tenure'], bins=30, kde=True)
```

10. MONTHLY CHARGES



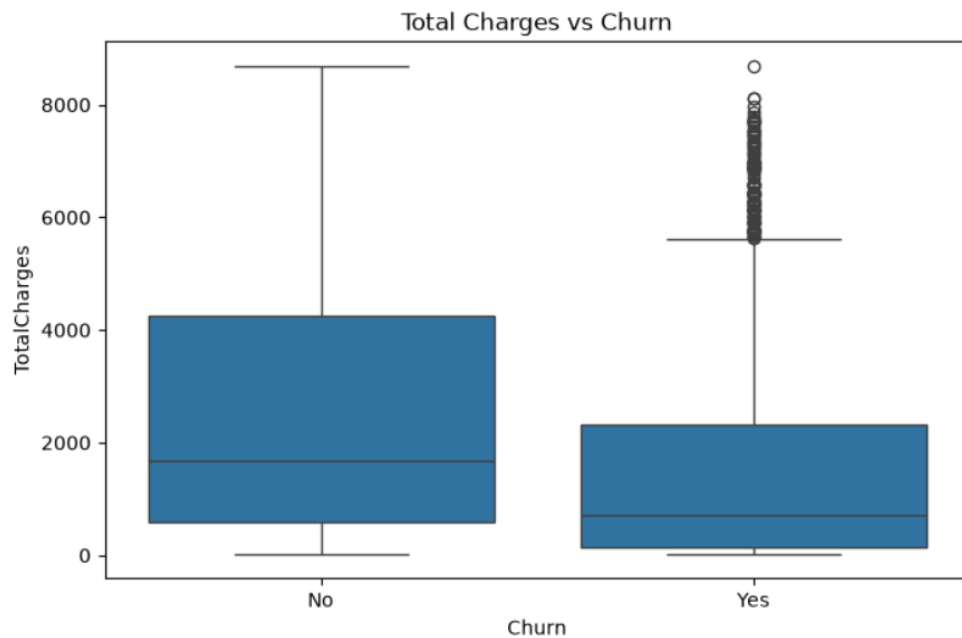
```
sns.histplot(df['MonthlyCharges'], bins=30, kde=True)
```

11. TENURE VS CHURN



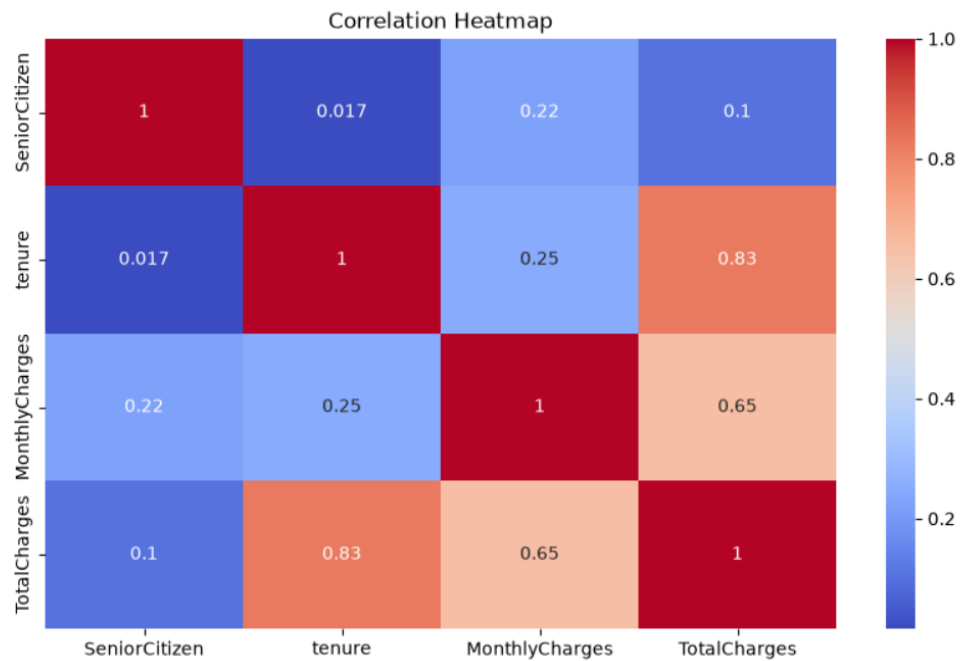
```
sns.boxplot(x='Churn', y='tenure', data=df)
```

12. TOTAL CHARGES VS CHURN



```
sns.boxplot(x='Churn', y='TotalCharges', data=df)
```

13. CORRELATION HEATMAP

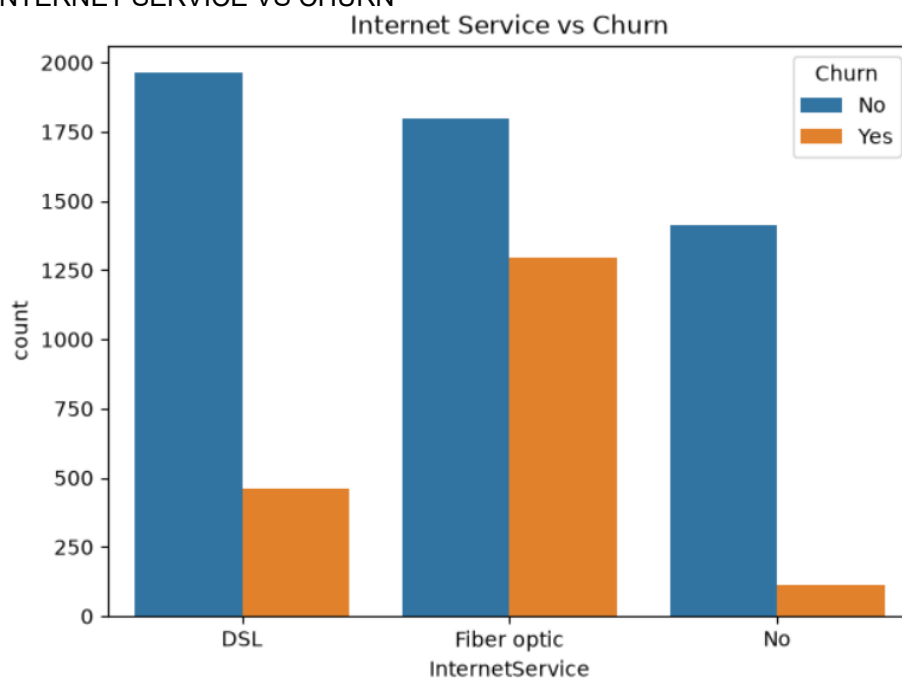


15. INTERNET SERVICE VS CHURN

n* 11 11 2 2

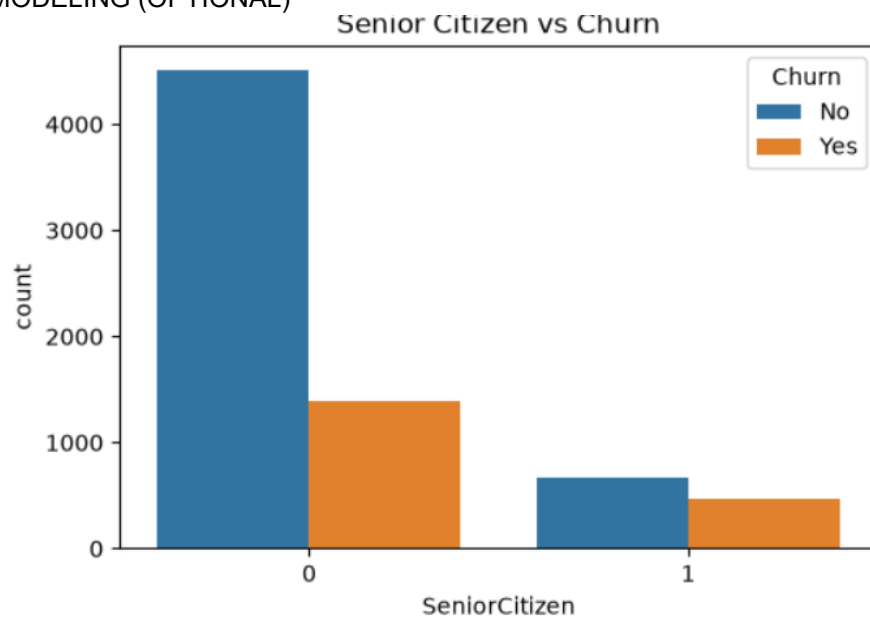
```
sns.heatmap(df.select_dtypes(include='number').corr(), annot=True)
```

14. INTERNET SERVICE VS CHURN



```
sns.countplot(x='InternetService', hue='Churn', data=df)
```

15. MODELING (OPTIONAL)



```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

df['Churn'] = df['Churn'].map({'Yes':1, 'No':0})
df = pd.get_dummies(df, drop_first=True)

X = df.drop('Churn', axis=1)
y = df['Churn']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

